I Main Idea

- Little / no tuning
- Explore rapidly - avoid computationally expensive strategies

II Denseness

(Informal) Create a sequence of points that get's arbitrarily close to any point in the configuration

A set U is dense in V if $cl(U) = V$

closed set

'Dense sequence':

Underlying set of points (i.e unordered) is dense

II Rapidly Explore Random/Dense Trees

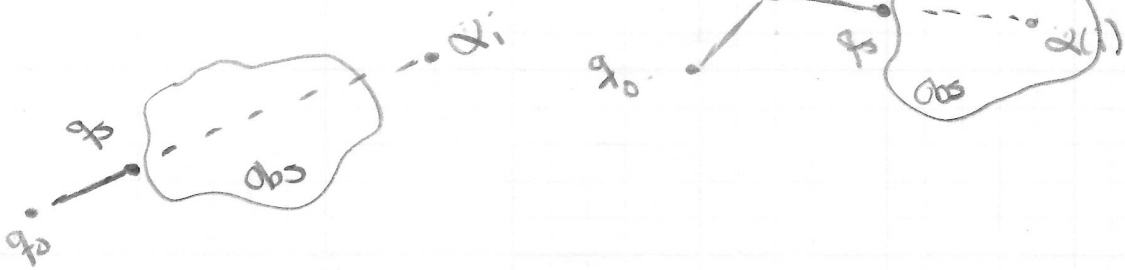
1) Randomly select points

2) Make a tree of nodes where you append new nodes to their most adjacent nodes.

Not an optimal solution, but it is feasible

A) Problem - Obstacles

If random node is within or the line goes through an object, find the nearest node that is in the direction of the random node and is outside the obstacle



→ We also have to make sure we don't insert the same node twice.

B) Goal Bias

Every so often greedily set the next q to sample as the goal.

1) Notes

Bias too strong: Greedy (reduce exploration)

Bias too weak: Search (reduce exploitation)

C) RDT Balanced Bidirectional Search

Explore/Bias from q_0 & q_{goal}

↳ Make 2 trees where one is initialized with q_{goal} and the other with q_0.

$T_a(q_0)$ $T_B(q_g)$ ← Trees

Begin exploring w/ T_a with no ties (fully explore)

Begin exploiting w/ T_b to try and connect q_n in T_b with the nearest node from T_a.

↳ (Fully Exploit)

If $|T_b| > |T_a|$ swap T_a & T_b

There are other approaches

If trees connect → STOP

III RRT - A Procedure-based View

1) $x_{rand} \leftarrow$ Sample(X_s)

2) $x_{near} \leftarrow$ Nearest(x_{rand}, T)

3) $N_{ear} \leftarrow$ Near$_{\rho,d}(x_{rand}, T)$

4) $x_n \leftarrow$ Steer$_\eta(x_g, y)$

5) $T \leftarrow$ InsertNode(x_n, x_p, T)

⋮

Formulates into a sequence of steps

IV RRT Properties and Further Development

A) Versitile

1) Escape local minima

2) Find topology to traverse

B) Properties

1) Dense w/ probability 1

2) Probabilitically complete:

as $t \to \infty$ through iterations probability of finding ~~qq~~ is 100%.

3) Optimal w/ shortest path w/ probability 0

C) Future Considerations

1) Properties of sampling

2) How to find optimal path

3) How do you define closest point in configuration/state space

4) How to create a new edge